Research and Adaptation of Components from the Persistent OS Phantom for Genode

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June 2026

Relevance

- Orthogonal persistence has been developing since the 1980s
- New equipment spurs new research
- The current situation requires a research

Objective

Current implementation:

- Test systems exist
- Embedded systems
- No permanent physical access is required

Goal: Explore options for ensuring data security

Data Persistence

Persistence the period of time during which an object is usable[1]

Today's dominant model — **two-level store** (non-persistent):

- Data is either *active* (RAM) or *stored* (disk)
- Programmer manually moves data between levels

Consequences:

- Data loss
- I/O boilerplate (≈30% of code)
- Frequent initialization and configuration

[1] M. P. Atkinson, P. J. Bailey, K. J. Chisholm, W. P. Cockshott, and R. Morrison, "Ps-algol: A language for persistent programming," 1983.

Orthogonal Persistence

Alternative: the **environment** manages persistence

Principles of **orthogonal persistence**[2]:

1. **Independence**
2. **Data type orthogonality**
3. **Identification**

Result: No two-level store model's drawbacks and improved disk's bandwidth utilization

[2] R. Morrison and M. P. Atkinson, "Persistent Languages and Architectures," 1990.

PhantomOS

PhantomOS[3] is a non-Unix like OS implementing orthogonal persistence.[4]

- Phantom Virtual Machine (PVM): bytecode language VM
- Simplified POSIX layer (non-persistent)
- Periodical snapshots (via **Copy-on-Write**)
- Changed pages written to disk (incremental)
- Programs are temporarily suspended (snapshot preparation)

[3] `http://phantomos.org/`
[4] D. Zavalishin, PhantomOS, 2010.

Genode

PhantomOS ported to the **Genode** OS C++ framework[5]:

- Proven microkernels (NOVA, seL4, Fiasco.OC)
- Capability-based security
- Minimal TCB

[5] `https://genode.org/`

Snapper

Snappser[6] is a file-system-based transactional snapshot mechanism[7]

Previous approach ("superblock"): snapshots stored as monolithic objects.

+ Fast
− No fault tolerance, no storage recovery

Snapper 2.0:

- Configured generations number
- Unchanged pages **linked**, not copied across generations
- ext4 + fsck for crash recovery
- Integrity checks (hashes per file)
- Configurable redundancy policy

[6] `https://github.com/rumenmitov/snapper`

[7] R. Mitov, "Snapper (v2): A PhantomOS snapshot mechanism," FOSDEM 2026. `https://fosdem.org/2026/schedule/event/DJ8YVT-practical-persistence/`

Problem Statement

Problem:

- Snapshots contain **sensitive data**
- **Plaintext** snapshots are stored on the drive

Consequences:

- Data can be **read directly**
- Ability to **restore the complete state**

Immediate decision's challenges:

- **No prior research** on this for PhantomOS
- **No ready-made solution** exists in Genode

Solution: to implement a mechanism for the secure storage of snapshots.

Environment & Requirements

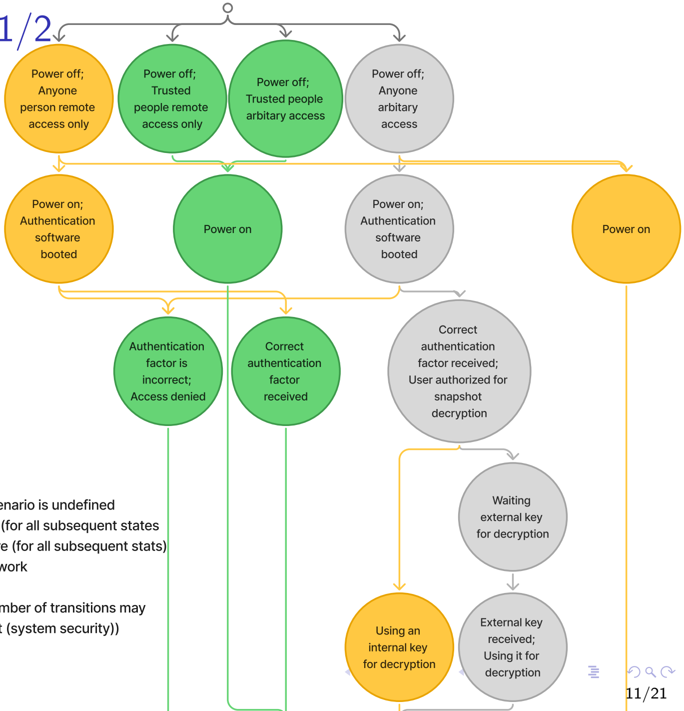
No unauthorized access to data

Requirements:

1. Mandatory authorization
2. Key stored on an **external USB drive**
3. Encryption of snapshots (AES, block-level)
4. Only encrypted data written to disk
5. Acceptable performance overhead ($\leq 2\times$)

Requirements 1 & 2 combined: the USB drive contains the key and serves as the authentication factor.

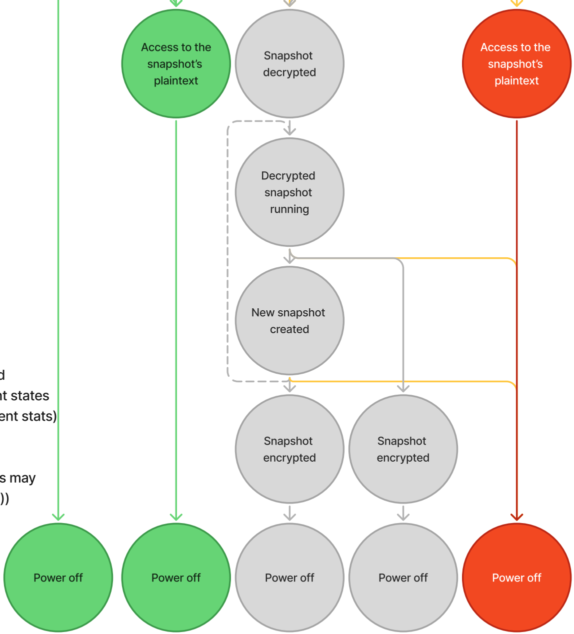
Usage scenarios 1/2



Legend:

- System state (computer - user)
- Note to the state
- In this state, the security of the scenario is undefined
- In this state the scenario is secure (for all subsequent states
- In this state the scenario is insecure (for all subsequent stats)
- This scenario is considered in our work
- ↓ Transition between states
- ↓ Transition between states (The number of transitions may vary and does not affect the final result (system security))

Usage scenarios 2/2



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Implementation: se_vault

file_vault based on **Tresor** library:

- Tresor: AES encryption on 4K blocks, SHA-256 integrity, CoW atomicity
- file_vault: state orchestrator providing a secure FS session

Implemented **se_vault**[8] based on **file_vault**[9]:

- Headless operation
- Passphrase replaced by **USB key file**
- rump_fs/ext2 replaced by **lwext4/ext4**
- Added **fsck** crash recovery support with fsck
- Direct block session (skip image file creation)
- All block sizes standardized to 4K
- Graceful shutdown sequence added

[8] https://github.com/VoronM1522/genode/tree/se/repos/gems/src/app/se_vault

[9] https://github.com/genodelabs/genode/tree/sculpt-25.04/repos/gems/src/app/file_vault

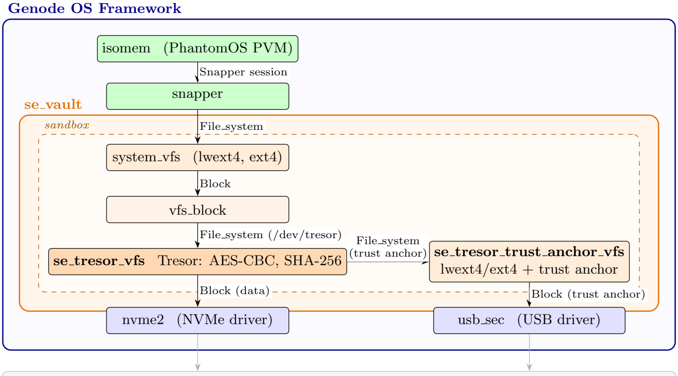
Related Fixes & Adaptations

Functionality patches:

- lwext4: `ftruncate` and `sync`
- isomem: power-off button callback and graceful shutdown
- snapper: client counter and graceful shutdown
- vfs: client counter, graceful shutdown and clean unmount

Optimization: using a "Block" session directly
reduced initialization time from $\approx$20 min to $\approx$1 min.

System Architecture



QEMU `-cpu host`, virtual NVMe (PCIe), USB passthrough

Podman container (USB flash passthrough: `--device /dev/bus/usb/...`)

Host Linux
- NVMe disk (btrfs): `block0.raw`, `test.5gb.raw`
- USB flash (ext4): trust anchor key, initialization flag

HW: NVMe SSD, USB 2.0 flash drive

First run configuration

Environment: QEMU with `-cpu host` (enables AES/SHA hardware acceleration).
2 GB image for isomem, 5 GB image for se_vault (4 GB user FS).

Functional testing

Functional tests (all passed):

- Normal boot / shutdown / restore cycle
- Encrypted image cannot be mounted without key
- Key/hash replacement attack — superblock not detected
- Crash recovery via fsck

Performance: first snapshot creation and restore, 5 runs each.

Performance testing

Functional tests (all passed):

- Normal boot / shutdown / restore cycle
- Encrypted image cannot be mounted without key
- Key/hash replacement attack — superblock not detected
- Crash recovery via fsck

Performance: first snapshot creation and restore, 5 runs each.

Results

The measurements are not statistically significant, but they provide an idea of the costs involved in the mockup implementation.

Operation	Without enc.	With enc.	Overhead
Snapshot creation	2.10 s	51.00 s	$\approx 24\times$
Snapshot restore	2.50 s	27.30 s	$\approx 11\times$

Root cause: **known Tresor library limitation** — not optimized for multithreading. Acknowledged by the author; unresolved.

Evaluation

4 of 5 requirements met.

- ✓ Mandatory authorization (USB key)
- ✓ Key on external medium
- ✓ AES block-level encryption
- ✓ Only encrypted data on disk
- $\times$ Overhead $\leq 2\times$ *(Tresor limitation)*

Security gain: snapshot cannot be mounted or read without the key.
Integrity: SHA-256 at block level (Tresor) + hash at FS level (Snapper).

Conclusion

Adapted **file_vault** → **se_vault** for encrypted snapshot storage in PhantomOS on Genode.

First security research specifically for PhantomOS.
No analogous component existed prior to se_vault.

Future work:

- Fix Tresor performance (or switch to FS-level encryption via OpenSSL)
- Encrypt the isomem block device using se_vault
- Investigate merging Snapper and Tresor snapshot concepts
- Bring the PoC implementation to an industrial-scale level